Supplementary Material: Auditable Alarm Diagnosis: Evidence-Role Separation and Risk-Bounded Selective Calibration for Industrial Fault Disposition

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Paired Retrieval Differences

The independent resampling unit is the query ($n = 49$); the 1225 query–case grades are not treated as independent samples. All methods were scored on the same frozen automated model-generated assay.

Table 1: Complete query-bootstrap paired differences on the 49-query automated model-generated retrieval assay. Values are candidate minus Structured QAKB with 95% intervals. The assay is not human ground truth.

Method minus Structured	Strict@1	Strict MRR	nDCG@5	Broad@1	Broad MRR
DTW	-0.0510 [-0.1122, 0.0102]	-0.0358 [-0.0772, 0.0034]	-0.0150 [-0.0323, 0.0001]	-0.0102 [-0.0306, 0.0000]	-0.0051 [-0.0204, 0.0051]
Embedding	-0.0102 [-0.0714, 0.0510]	-0.0160 [-0.0580, 0.0230]	-0.0123 [-0.0341, 0.0077]	0.0204 [0.0000, 0.0510]	0.0119 [0.0000, 0.0306]
Multiscale	-0.0510 [-0.1122, 0.0102]	-0.0355 [-0.0774, 0.0039]	-0.0159 [-0.0348, 0.0001]	-0.0102 [-0.0306, 0.0000]	-0.0051 [-0.0204, 0.0051]
Two-branch hybrid RRF	-0.0204 [-0.0714, 0.0408]	-0.0097 [-0.0395, 0.0201]	-0.0055 [-0.0158, 0.0043]	-0.0102 [-0.0306, 0.0000]	-0.0034 [-0.0153, 0.0051]
Full hybrid RRF	0.0000 [-0.0612, 0.0612]	0.0036 [-0.0267, 0.0349]	-0.0011 [-0.0140, 0.0108]	-0.0102 [-0.0306, 0.0000]	-0.0051 [-0.0187, 0.0034]

Expert-Graded nDCG@5 Computation

The expert-graded nDCG@5 of 0.918 (95% CI 0.883–0.947) reported in the main text is computed on the blinded 160-pair expert subset, which covers the structured channel's query–case pairs. Each of the two domain experts assigned an ordinal relevance grade under the same three-level scale as the automated assay (2: close event analogue; 1: broadly relevant analogue; 0: off-target). For query q, with the structured channel's retrieved cases in ranked order and expert grade $g_i(q) \in \{0, 1, 2\}$ for the case at rank i, the discounted cumulative gain at rank 5 is

$$\mathrm{DCG@5}(q) = \sum_{i=1}^{5} \frac{2^{g_i(q)} - 1}{\log_2(i+1)},$$

and nDCG@5 normalizes by the ideal DCG obtained by sorting the available grades in descending order:

$$\mathrm{nDCG@5}(q) = \frac{\mathrm{DCG@5}(q)}{\mathrm{IDCG@5}(q)}.$$

Queries whose ideal gain is zero are handled by the frozen convention recorded in the released replay artifacts. The reported value is the mean over queries; the interval uses the frozen query-level bootstrap, matching the automated assay endpoints. Because the expert subset graded only the structured channel's pairs, this quantity validates the structured channel's ranking against human relevance and does not support cross-method re-ranking under expert grades.